Course structure of B.Tech (Civil Engineering) for batch 2024-28

Semester 5	OE		Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3
	HSC		Engineering Economics	3	0	0	3	3
	PE		Program Elective 1	3	0	0	3	3
	PC		Hydrology and Water Resources	3	0	2	5	4
	PC		Foundation and Geotechnical Applications	3	0	2	5	4
	PC		Design of RCC Structures	3	0	0	3	3
	PC		Computer Aided Design Drawing Lab - I	0	0	2	2	1
				18	**0**	**6**	**24**	**21**
Semester 6	OE		Open Elective 3 (From Other Department of FoET)	3	0	0	3	3
	PE		Program Elective 2	3	0	0	3	3
	PE		Program Elective 3	3	0	0	3	3
	PC		Design of Steel Structures	3	0	0	3	3
	PC		Computer Aided Design Drawing Lab - II	0	0	2	2	1
	PC		Environmental Engineering	3	0	2	5	4
	PC		Highway and Traffic Engineering	3	0	2	5	4
				18	**0**	**6**	**24**	**21**

Course structure of B.Tech (Civil Engineering) for batch 2024-28

Semester 7	Pro		Summer Internship	0	0	0	0	2
	OE		Open Elective 4 (From Other Department of FoET)	3	0	0	3	3
	PE		Program Elective 4	3	0	0	3	3
	PE		Program Elective 5	3	0	0	3	3
	PC		Estimation Costing Contracts and Valuations	3	1	0	4	4

					DESIGN OF STEEL STRUCTURES					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the design the various types of bolted connections used in steel structures
2. To study various types of welded connections and concept of design of tension members
3. To understand the concepts and procedure for design of the simple and composite columns including column base and gusseted base.
4. To understand procedure to design laterally supported and unsupported steel I beams

UNIT I : BOLTED CONNECTIONS **10 Hrs.**

Introduction to steel structures and steel code provisions, scope of the subject, types of connections . Bolted connections: types bolts, specification, permissible values, Butt and lap connections of steel plates or flats, numerical examples. Beam to beam connections numerical examples. Beam to column connections: Connections subjected axial load and twisting moments, numerical examples. Beam to column connections: Connections subjected axial load and bending moments, numerical examples

UNIT II : WELDED CONNECTIONS AND TENSION MEMBERS **10 Hrs.**

Introduction to welded connections: Butt and lap welded connections of steel plates or flats, numerical examples. Beam to column welded connections: Connections subjected axial load and twisting moments, numerical examples. Beam to column welded connections: Connections subjected axial load and bending moments, numerical examples. Design of tension members using angle, double angle, T and L rolled steel members, lug angle, numerical examples

UNIT III : COMPRESSION MEMBERS AND COLUMN BASES **10 Hrs.**

Design of compression members using I and Channel sections, Compound column design using double channel sections, numerical examples. Compound column design using lattice connections, numerical examples. Design of column splice of various types using numerical examples. Design of column base and gusseted base concept and numerical examples

UNIT IV : LATERALLY SUPPORTED AND UN-SUPPORTED BEAMS **12 Hrs.**

Design of beams using I sections, design principles, code provisions. Design of laterally supported beams subjected to uniformly distributed loads, design procedure, design numerical examples. Design of laterally un supported beams subjected to uniformly distributed loads, design procedure, code provisions design numerical examples.

TOTAL HOURS:42 Hrs.

COURSE OUTCOMES

CO-1: Design the bolted concentric and eccentric connections
CO-2: Design the welded concentric and eccentric connections
CO-3: Design the tension members
CO-4: Design the simple and compound compression members.
CO-5: Design column base and gusseted base
CO-6: Design laterally supported and unsupported beams.

TEXT-BOOK AND REFERENCE BOOKS:

1. K.S. Sai Ram, "**Design of Steel Structures",** Pearson Publication, Delhi
2. Subramanian N., "**Design of Steel Structures**", Oxford University press, New Delhi.
3. Duggal S.K., "**Limit State method of design of steel structures**", Tata McGraw-Hill, New Delhi
4. Arya A.S and Ajmani J.L, "**Design of Steel Structures**", Nemchand Bros., Roorkee.
5. "**IS 800-2007: General construction in steel-Code of practice (third revision)**", Bureau of Indian Standards, New Delhi.
6. Steel Table

					COMPUTER AIDED DESIGN DRAWING LABORATORY - II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To illustrate bolted and welded connections
2. To illustrate lays connections and column base connections
3. To understand and draw beam connections.

LIST OF EXPERIMENTS

1 Understand basic commands of drafting software
2 Beam to beam bolted and welded connections
3 Beam to column bolted and welded connections
4 Beam to beam welded connections
5 Beam to column welded connections
6 Bolted or welded Butt and lap connections
7 Column splice connections
8 Single and double lacing or Batten connections in compound columns
9 Column base and gusseted base connections
10 Simply supported or cantilever truss drawing
11 Understand basic commands of FEM software
12 Analysis of Truss using FEM Software
13 Analysis of Industrial building frame using FEM software

COURSE OUTCOMES

CO-1: Explain basic commands of drafting software
CO-2: Draft beam to beam or Beam to column connections
CO-3: Construct the butt, lap and column splice connections
CO-4: Draw lacing, batten and column base connections in compound columns
CO-5: Construct drawings of simply supported or cantilever Truss
CO-6: Analyse Truss or industrial building frame using FEM software

TEXT-BOOK AND REFERENCE BOOKS:

1. N Krishnaraju, "**Structural Design and Drawing**", University press
2. C.S.Krishna Murthy & Rajiv S. – Computer Aided Design, software & Analytical tools - Narasha publishing house India.
3. K.S. Sai Ram, **Design of Steel Structures**, Pearson Publication, Delhi
4. Subramanian N., "**Design of Steel Structures**", Oxford University press, New Delhi.

Teaching Scheme					Environmental Engineering					
					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	4	5	25	50	25	--	--	100

COURSE OBJECTIVES

1. Understanding the various aspects of water - sources, quality, demands
2. Design of a Water Treatment Plant
3. Understanding the qualitative and quantitative aspects of wastewater
4. Design of a Wastewater Treatment Plant

UNIT I : DRINKING WATER **12 Hrs.**

Water cycle, Sources of water, Water quality standards, Impact of water quality on - human health, agriculture, materials, etc. Water Demands types and estimation, Water Budget, Population Forecasting Methods.

UNIT II : WATER TREATMENT **16 Hrs.**

Design of Screens, Types of settling and settling tests, Design of Sedimentation Tank, Coagulation: mechanisms of coagulation, Mixing and Flocculation, Design of clariflocculator, Filtration: Slows sand filters, rapid sand filters, Water Softening, Water Disinfectioning, Action plan for efficient plant operation

UNIT III : WASTEWATER **12Hrs.**

Classification of wastewater, Composition of wastewater, Characterization of wastewater (physical, chemical and biological), basics of sewage quality assessment, sources of wastewater, estimation of wastewater quantity.

UNIT IV : WASTEWATER TREATMENT **16Hrs.**

Significance of wastewater treatment, objectives of a sewage treatment plant, classification of treatment processes, design of a sewage treatment plant - Screening, Grit Removal Basin, Skimming Tank, Sedimentation Tank, Secondary Treatment of sewage through (i) Biological Filtration of Sewage (ii) Activated Sludge Process, Disinfection of sewage and sludge, methods of disposal of treated Sewage along with their standards**,** Action plan for efficient plant operation.

TOTAL HOURS 56 Hrs.

COURSE OUTCOMES

CO1 – Estimate the present and future needs of water of a city
CO2 – Design a Water Treatment Plant
CO3 – Propose measures for efficient functioning of a water treatment plant
CO4 – Classify the wastewater and quantify its pollution load
CO5 – Design a Wastewater Treatment plant
CO6 – Propose measures for efficient functioning of a wastewater treatment plant

TEXT-BOOK AND REFERENCE BOOKS:

1. S. K. Garg, "**Water Supply Engineering (Environmental Engineering)**", Khanna Publishers, 2016
2. S K Garg, "**Sewage Disposal and Air Pollution Engineering**", Khanna Publishers, 2016 GS Birdie, JS Birdie, Water Supply and Sanitation Engineering, Galgotia Publishing Ltd
3. HS Paevy, DR Rowe, G Tchobanoglous, "**Environmental Engineering**", McGraw Hill
4. Metcalf and Eddy, "**Water and Wastewater Engineering**", McGraw Hill
5. Sawyer, C.N. and McCarty, P.L., and Parkin, G.F., "**Chemistry for Environmental Engineers**", 4th Edn. McGraw Hill, New Delhi, 1994.
6. Benefield, Judkins and Weand, "**Process Chemistry for Water and Wastewater Treatment**", Prentice Hall

Teaching Scheme					Environmental Engineering (Laboratory)					
					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	0	0	0	50	50	100

COURSE OBJECTIVES

5. Understanding the sampling procedures for water and wastewater collection
6. Estimating the pollutants present in the Water through experimentation
7. Estimating the pollutants present in the Wastewater through experimentation
8. Analyze the Water and Wastewater quality based on the respective standards

DETAILS OF LABORATORY PRACTICALS:

1. Determination of pH, Turbidity
2. Determination of Alkalinity and Acidity
3. Determination of Total Solids, Total Dissolved Solids, Total Suspended Solids
4. Determination of Total Hardness and Calcium Hardness
5. Determination of Chlorides
6. Determination of Coagulant Dosage by Jar Test
7. Determination of Residual Chlorine
8. Determination of Dissolved Oxygen
9. Determination of BOD
10. Determination of COD
11. Project – 1 Assessment of the drinking water of the university campus / residential Building
12. Project – 2 Assessment of the treated wastewater of the university campus

COURSE OUTCOMES

CO-1: Estimate the concentration of a pollutant present in the water
CO-2: Analyze the quality of water based on water quality standards
CO-3: Propose appropriate measures for improving the water quality
CO-4: Estimate the concentration of a pollutant present in the wastewater
CO-5: Analyze the quality of wastewater based on wastewater quality standards
CO-6: Propose appropriate measures for improving the wastewater quality

TEXT-BOOK AND REFERENCE BOOKS:

1. Standard Methods for The Examination of Water and Wastewater (American Water Works Association), ISBN-9780875532875
2. Indian Standards (IS: 3025)
3. Indian Standards (IS: 10500-2012)

					Highway and Traffic Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	4	5	25	50	25	--	--	100

COURSE OBJECTIVES

1. To give an overview about highway engineering with respect to planning and alignment.
2. To know the importance of geometric design.
3. To learn the procedure and method of pavement design as per IRC.
4. To study the various traffic parameters and its estimation and design.

UNIT I : PRINCIPLES OF TRANSPORTATION ENGINEERING **12 Hrs.**

Importance of transportation- Different modes of transportation and comparison-Characteristics of road transport- Road type's classification- road patterns- various road development plans in India- factors involved in alignment of new road link- planning surveys- Indian Roads Congress Guidelines.

UNIT II : HIGHWAY GEOMETRIC DESIGN **16 Hrs.**

Importance of geometric design- Terrain classification- Design speed- Factors affecting geometric design, Cross sectional elements-Camber- width of pavement- Sight Distance design- Stopping sight distance- Overtaking sight distance overtaking zones- Examples on SSD and OSD- Sight distance at intersections- Horizontal alignment-Radius of Curve- Super elevation – Extra widening- Transition curve and its length- Setback distance – Examples, Vertical alignment-Gradient summit and valley curves.

UNIT III : PAVEMENT MATERIALS AND PAVEMENT DESIGN **16 Hrs.**

Desirable properties and different test procedure of pavement materials (tests for soil, aggregate, bitumen etc.- Road construction methods- PAVEMENT DESIGN- Pavement types- component parts of flexible and rigid pavements and their Functions- design factors- ESWL and its determination- Flexible pavement- Design of flexible pavements as per IRC37-2018. Rigid pavement- Westergaard's equations for load and temperature stresses- Design of slab thickness only as per IRC: 58-2015. Different types of Flexible and Rigid pavement failures and pavement management methods

UNIT IV : TRAFFIC ENGINEERING **12 Hrs.**

Various traffic parameters and its importance in the safe and efficient operation of highway. Traffic studies on flow and speed, peak hour factor, accident study, statistical analysis of traffic data; Microscopic and macroscopic parameters of traffic flow, fundamental relationships; Traffic signs; Signal design by Webster's method; Types of intersections; Highway capacity Introduction to Intelligent transportation- Different methods of ITS technologies developed for effective traffic operation and management.

TOTAL HOURS: 56 Hrs.

COURSE OUTCOMES

CO-1: Describe various factors considered for planning and alignment of Highway.
CO-2: Understand different geometric parameters and its importance in design.
CO-3: Compute geometric parameters and can design highway components as per requirement.
CO-4: Conduct different pavement materials tests and check suitability of construction material.
CO-5: Design flexible and rigid pavement as IRC.
CO-6: Create modern safe and efficient traffic network system.

TEXT-BOOK AND REFERENCE BOOKS:

1. L.R. Kadiyali," **Highway Engineering**", Khanna Publishers, New Delhi.
2. Dr. S.K. Khanna and Dr. C.E. G. Justo, "**Highway Engineering**", Nem Chand & Bros., Roorkee.
3. S.K. Sharma, Principles, "**Practice and Design of Highway Engineering**", S. Chand & Co., New Delhi.
4. L.R. Kadiyali, "**Traffic Engineering and Transport Planning**", Khanna Publishers, New Delhi.
5. K P Subramanium, "**Transportation Engineering**", Scitech Publications, Chennai
6. "**IRC: 37-2018 Guidelines for Design of flexible Pavements**", IRC.
7. "**IRC: 67 Code of Practice for Road Signs**", IRC, New Delhi – 2001.
8. "**IRC: 58 - 2015: Guidelines for the Design of Plain Jointed Rigid Pavements for Highways**", IRC.

					Highway and Traffic Engineering Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To know different test performed for aggregate and understand the procedure.
2. To know different test performed for Soil and understand the procedure.
3. To know different test performed for Bitumen and understand the procedure.
4. To know different traffic study methods

LIST OF EXPERIMENTS

1 Introduction to Highway Engineering Laboratory Equipment (Introduction class)
2 Aggregate crushing Test
3 Aggregate Impact Test
4 Flakiness Index and Elongation Index Test for Aggregate
5 Los Angeles Abrasion Test / Deval Abrasion Test
6 Specific gravity and Water Absorption test for Aggregate
7 Penetration test for Bitumen
8 Softening point test for Bitumen
9 Ductility test for Bitumen
10 Flash and Fire Point test for Bitumen
11 Specific gravity test for Bitumen
12 Viscosity Test for Bitumen
13 Marshall stability test on Bitumen mix
14 Conduct classified traffic volume study and spot speed study on busy rural highway or urban street during peak hour to
obtain the peak hour flow and design speed of a selected road section.

COURSE OUTCOMES

CO-1: Identify engineering properties of aggregate by performing laboratory tests necessary for highway construction.
CO-2: Demonstrate engineering properties of soil by performing laboratory tests necessary for highway design and construction
CO-3: Determine the grade & properties of bitumen by performing laboratory tests necessary for highway construction
CO-4: Design Bitumen Mixes
CO-5: Analyze the pavement material test results and suitability for construction purpose.
CO-6: Create smart materials which can be used for cost effective construction of road.

TEXT-BOOK AND REFERENCE BOOKS:

1. L.R. Kadiyali, **Principles and Practices of Highway Engineering**, Khanna Publishers, 2009
2. MoRTH (2013) **Specification for Road and bridge works** (5th revision)
3. S. K. Khanna, C. E. G. Justo, A Veeraragavan, **Highway Engineering**, Khanna Publishers.
4. IRC: 37-2018 **Guidelines for Design of flexible Pavements**, IRC.
5. IRC: 58 - 2015: **Guidelines for the Design of Plain Jointed Rigid Pavements for Highways**, IRC.